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Galaxies in Collision

A dark night & a big scope produced fantastic views of these seven interacting pairs.

Halton Arp's *Atlas of Peculiar Galaxies* (1966) and Boris Vorontsov-Velyaminov's *Atlas and Catalogue of Interacting Galaxies* (1959) offer a sample of several hundred interacting and merging galaxies. Unlike symmetrical spiral and elliptical galaxies, or known irregular galaxies such as the Magellanic Clouds, many of these merging pairs display strange structures such as loops, narrow streamers, and rings.

When the Arp and Vorontsov-Velyaminov catalogs were published, the physical processes involved in shaping these galactic deformities were poorly understood. But the carnage clearly resulted from the interaction, and the atlases provided intriguing targets for future investigations. During the past two decades, space and ground-based telescopes have studied the dynamics of interacting galaxies across X-ray, ultraviolet, visual, infrared, and radio wavelengths, and the high-resolution of the Hubble Space Telescope (HST) cameras have captured the galactic mayhem in exquisite detail.

A 12- to 18-inch telescope will display subtle structure in many colliding galaxies. But through Jimi Lowrey's 48-inch f/4 telescope in West Texas, the views can rival a high-quality image. Under good observing conditions we routinely pick out jaw-dropping interaction features such as tidal tails, bridges, and starbursts.

The Penguin & Egg

Arp 142 (VV 316) shows the violent merger of NGC 2936, a highly disrupted spiral resembling a celestial penguin on the HST image at right, and an egg-like elliptical companion, NGC 2937. The "eye" of the penguin is marked by the nucleus of NGC 2936, and a remnant arm, ablaze with knots of massive blue stars, serves as the "beak." Gravitational forces pulled gas and dust out of the plane of NGC 2936, warped the disc, and triggered torrents of blue star-streams in the shredded "neck." NGC 2936 took the brunt of this cosmic train wreck, while the elliptical NGC 2937 survived the encounter relatively unscathed.

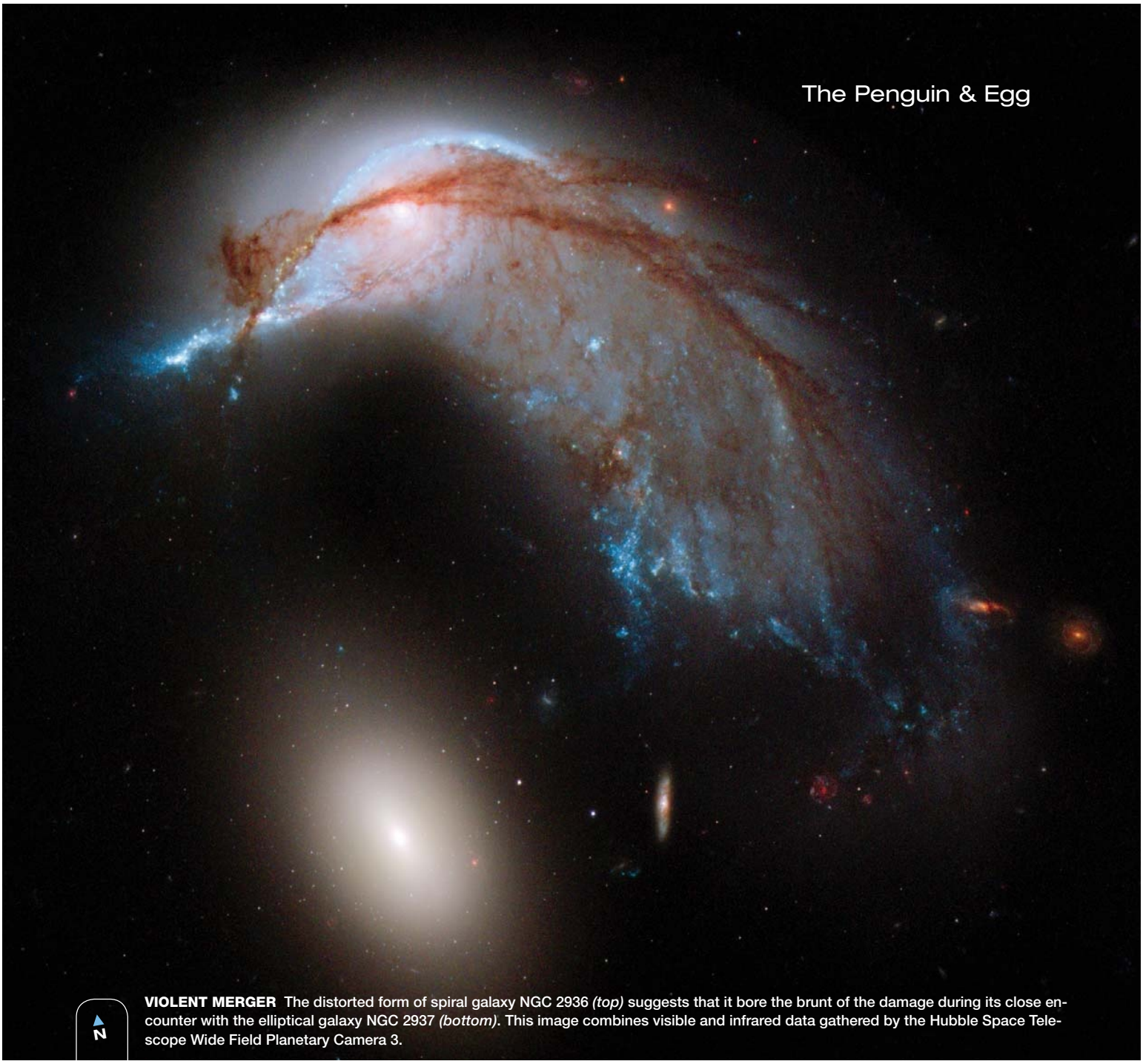
This small pair resides in Hydra, just southwest of its border with Sextans and Leo, at a distance of 320 million light-years. Viewed through the 48-inch, the surface brightness of NGC 2936 is highly irregular, showing a slightly curving body and sharply bending tail. The central region extends

The Vorontsov-Velyaminov Catalogs

Although Halton Arp's 1966 *Atlas of Peculiar Galaxies* is better known, Boris Vorontsov-Velyaminov (Sternberg Astronomical Institute, Moscow University) published a list of interacting galaxies some seven years earlier in *Soviet Astronomy*. The list consisted of the preliminary results of his study of the photographic plates taken during the first Palomar Observatory Sky Survey (POSS) in 1958. He followed this up with his 1959 *Atlas and Catalogue of Interacting Galaxies*, now known as VV Part I, which included 355 interacting galaxies, the majority of which he discovered on the POSS plates.

Vorontsov-Velyaminov issued a second catalog (known as VV Part II) in 1977, expanding the list of interacting galaxies by 497. Astrophysicist V. P. Arkhipova (Lomonosov Moscow State University) added an additional 1,162 galaxies originally cataloged by Vorontsov-Velyaminov to the *Morphological Catalogue of Galaxies* in 2001, expanding the numbered VV list to a total of 2,014.

• **FURTHER READING:** Both the Arp and Vorontsov-Velyaminov catalogs can be accessed via the NASA/IPAC Extragalactic Database (NED). See **https://is.gd/VVCatalog** for the VV catalog and **https://is.gd/ArpCatalog** for the Arp catalog.



VIOLENT MERGER The distorted form of spiral galaxy NGC 2936 *(top)* suggests that it bore the brunt of the damage during its close encounter with the elliptical galaxy NGC 2937 *(bottom)*. This image combines visible and infrared data gathered by the Hubble Space Telescope Wide Field Planetary Camera 3.

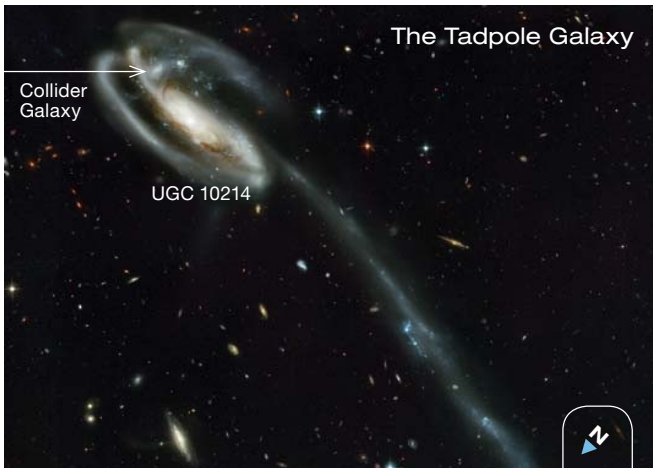
NASA / ESA / HUBBLE HERITAGE TEAM (STSCI / AURA)

30″ × 20″, tapering on the east end and punctuated by a conspicuous nucleus. A broad, low surface brightness tidal tail is attached on the west side. It curls south-southwest, gradually fading out due west of the intensely luminous but diminutive NGC 2937.

A 13th-magnitude star shines 1′ northwest of NGC 2937 and a ghostly 16.5-magnitude sliver (PGC 1237172) points northwest from the star. Arp called this object a "shred" ejected from NGC 2936, but its radial velocity implies the companion is in the foreground at only 70% the distance, so it isn't physcially related to the Penguin.

The Tadpole Galaxy

Arp 188 (VV 29, also cataloged as UGC 10214) is a severely deformed 13.7-magnitude spiral in Draco, a little over ½° southwest of 6.4-magnitude Theta (θ) Draconis. Fritz Zwicky photographed its huge tidal ribbon in the mid-1950s using the 200-inch at Palomar Observatory and remarked, "The system obviously has suffered some encounter with other galaxies which, however, cannot be located with certainty at the present time." The spectacular HST image on page 30 caught the small collider (a compact blue galaxy) partially hidden by a spiral arm. The close encounter, which occurred 100 to 200



▲ **THE INTERLOPER** The long plume of stars and dust trailing behind the spiral galaxy UGC 10214, the residue of its collision with a smaller galaxy, gives it the appearance of a tadpole racing across a pond. Close examination of UGC 10214's remaining structure reveals the guilty party, a compact, blue galaxy still entangled in the larger galaxy's spiral arms.

NASA / HOLLAND FORD (JHU) / ACS SCIENCE TEAM / ESA

million years ago, separated a major spiral arm from the galaxy and stretched it into a dramatic plume, rich in dust and gas. The galaxy's distance of 420 million light-years implies the Tadpole's tail unfurls across some 280,000 light-years!

In the 48-inch the main section of the galaxy extended 1.2′ × 0.6′ and encompassed an elongated central bar. The debris tail extending east was easily seen as a delicate, low surface brightness plume, nearly tripling the length of the galaxy. A couple of miniscule condensations sparkled, including one right at the tip. A 2003 HST study using the newly installed wide-field Advanced Camera for Surveys (ACS) concluded these clumps are super-massive clusters only 4 to 5 million years old. With time, they may evolve into dwarf companions in the Tadpole's halo.

The Heron Galaxy

Arp 84 (VV 48), the interacting pair NGC 5394 and NGC 5395 in Canes Venatici, is another stunner through the 48-inch at 610×. The flattened halo of 11.4-magnitude NGC 5395, which forms the Heron's "body" and "wings," spans 2.5′ × 1.0′ and envelops a blazing 30″ core. The most exciting feature is a well-defined, somewhat splotchy arm that wraps nearly 360° around the galaxy, forming a partial ring.

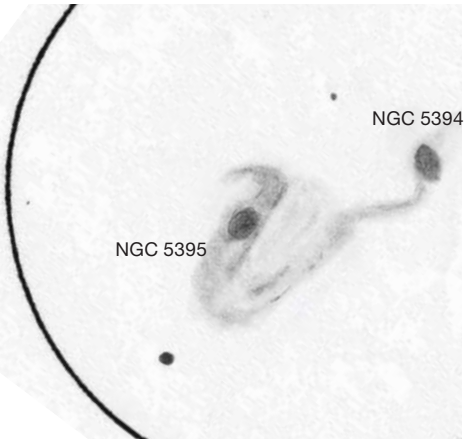
The arm emerges at the core's north edge, shoots directly south on the east side, and twists tightly backwards to the north along the west edge of the halo. A dark absorption lane clearly separates the luminous central region from this arm. An irregular offshoot on the north side includes a brighter patch.

NGC 5394, centered 2′ to the north-northwest, is a compact 13th-magnitude disc only 30″ in diameter with a strongly concentrated center and brilliant stellar nucleus. The Heron's neck, a long, thin tidal arm, was easily visible, hooking southeast and nearly touching the northwest arm of

Colliding Galaxies & Where to Find Them

Arp / VV Number	Object	Type	Mag(v)	Size	RA	Dec.	Nickname
Arp 142 / VV 316	NGC 2936	Irr	13.1	1.6′ × 0.9′	09^h 37.7^m	+02° 45′	The Penguin and Egg
	NGC 2937		13.7	0.8′ × 0.4′	09^h 37.8^m	+02° 45′	
Arp 188 / VV 29	UGC 10214	SBc pec	13.7	3.6′ × 0.8′	16^h 06.1^m	+55° 26′	Tadpole Galaxy
Arp 84 / VV 48	NGC 5394	SBb pec	13.0	1.7′ × 1.0′	13^h 58.6^m	+37° 27′	The Heron Galaxy
	NGC 5395	SAb pec	11.4	2.9′ × 1.5′	13^h 58.6^m	+37° 26′	
Arp 242 / VV 224	NGC 4676A	S0 pec	13.7	2.3′ × 0.5′	12^h 46.2^m	+30° 44′	The Mice
	NGC 4676B	SBa pec	13.6	2.0′ × 0.8′	12^h 46.2^m	+30° 43′′	
Arp 55 / VV 155	UGC 4881A	Double	14.5	0.9′ × 0.8′	09^h 15.9^m	+44° 20′	The Grasshopper
	UGC 4881B		15.1	—	00^h 15.9^m	+44° 20′	
Arp 244 / VV 245	NGC 4038	SBm pec	10.3	5.2′ × 3.1′	12^h 01.9^m	–18° 52′	The Antennae
	NGC 4039	SAm pec	10.6	3.1′ × 1.6′	12^h 01.9^m	–18° 53′	
Arp 81 / VV 247	NGC 6621	SBc pec	13.3	2.1′ × 0.9′	18^h 12.9^m	+68° 22′	Edward's Galaxy
	NGC 6622	E/S0 pec	11.9	0.6′ × 0.5′	18^h 13.0^m	+68° 21′	

Angular sizes and separations are from recent catalogs. Visually, an object's size is often smaller than the cataloged value and varies according to the aperture and magnification of the viewing instrument. Right ascension and declination are for equinox 2000.0.



BRIGHT EYE Even under difficult skies, with the right scope you might be able to pull out the luminous cores of these interacting galaxies, particularly the bright "eye" of the Heron, NGC 5394. This sketch was drawn using a 20-inch (505-mm) reflecting scope and a deep-sky video camera.

NGC 5395. The opposing arm ("head" and "bill") was spied as a tenuous smoke trail to the northwest.

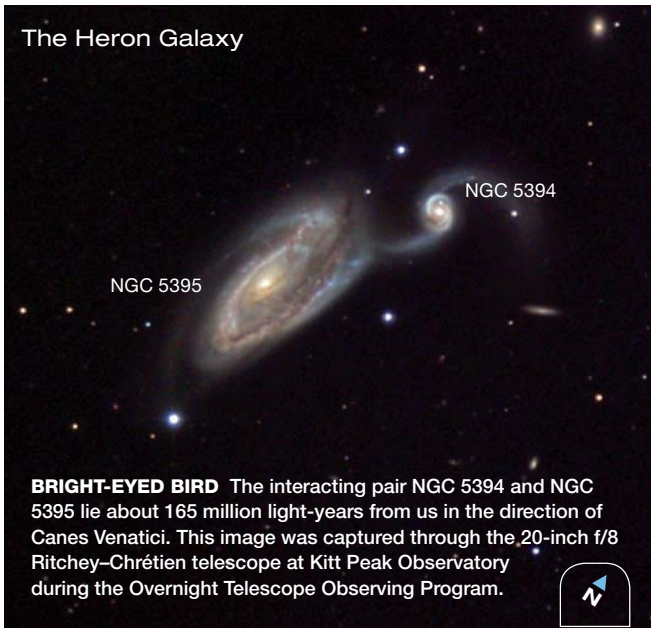
A near-infrared study published in the *Astronomical Journal* in October 2005 concluded, "This tightly wound structure [in NGC 5395], resembling a broken ring, supports the scenario of a strong interaction in the system, probably a collision with NGC 5394 crossing through the disk of NGC 5395." A spectroscopic investigation published in the *Monthly Notices of the Royal Astronomical Society* in 2015 by astronomers working with Calar Alto Legacy Integral Field Area (CALIFA) survey data reported key evidence of the earlier collision, with a flurry of star-forming activity a few hundred million years ago, as well as detailing an ongoing nuclear starburst in NGC 5394.

The Mice

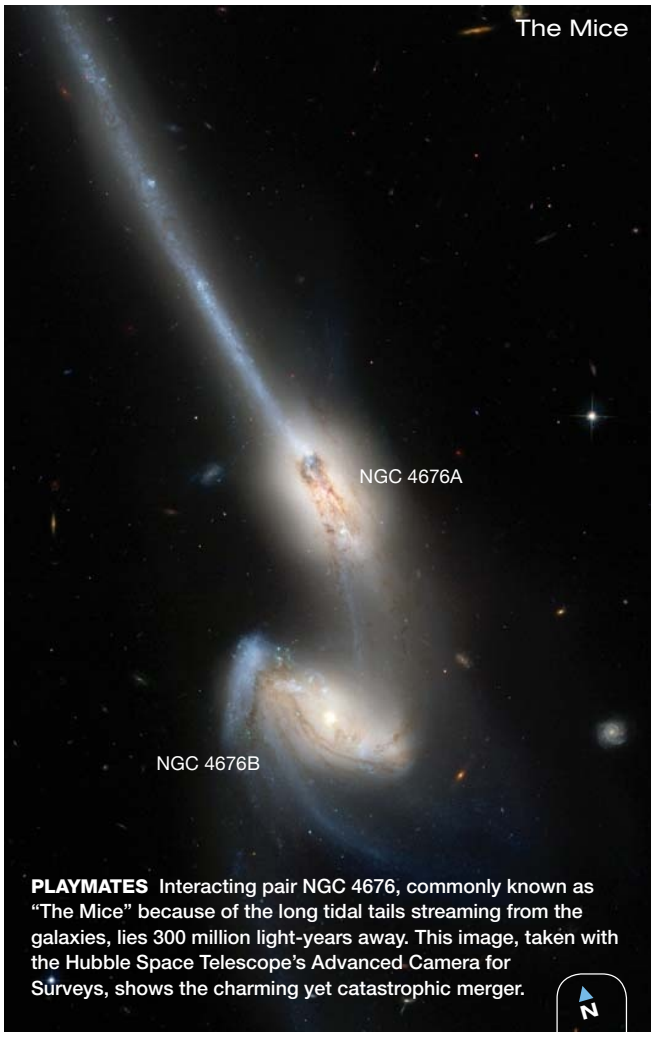
Vorontsov-Velyaminov dubbed the two interacting members of NGC 4676 that form **Arp 242** (VV 224) the "Playing Mice" based on their two oval bodies and long tails. This duo lies in northern Coma Berenices, just 1.5° south-southeast of the "Hockey Stick" galaxy NGC 4656, a well-known, highly distorted edge-on. But NGC 4676 lies far in the background at a distance of 300 million light-years in the outskirts of the rich Coma Galaxy Cluster (Abell 1656).

Astronomers consider the Mice the archetypal prograde merger of equal-mass disc galaxies — colliding with the same sense of rotation and orbital motion. In a seminal 1972 paper "Galactic Bridges and Tails," Alar Toomre (MIT) and Juri Toomre (University of Colorado) used computer simulations to trace the evolution at various time intervals and successfully reproduced the tidal features of the Mice. Joshua Barnes (University of Hawai'i) used a more sophisticated dynamical simulation in 2004 and concluded the collision occurred 170 million years ago with a final head-on encounter and coalescence destined in 540 million years.

NGC 4676A is a small, high surface brightness oval. Through the 48-inch, it extended 24″ × 16″. Its singular visual feature is a thin tidal tail shooting 80″ due north and dimming out just east of a 17.3-magnitude star. The surface brightness of this spike is the highest for any long tidal tail and is a good target for 12-inch or larger scopes under pristine skies.

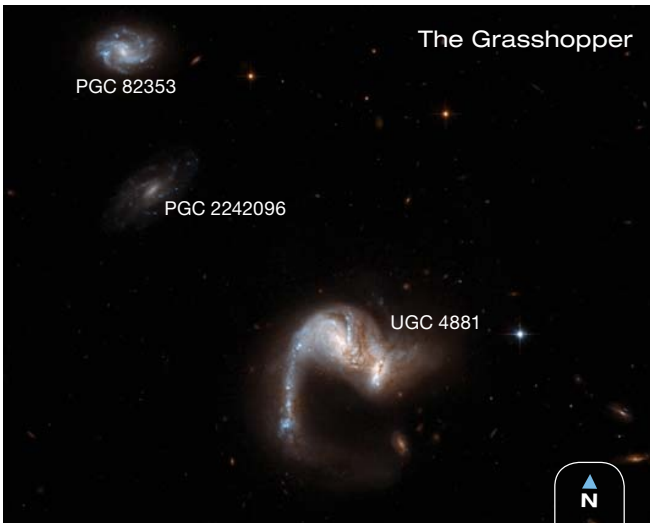


BRIGHT-EYED BIRD The interacting pair NGC 5394 and NGC 5395 lie about 165 million light-years from us in the direction of Canes Venatici. This image was captured through the 20-inch f/8 Ritchey–Chrétien telescope at Kitt Peak Observatory during the Overnight Telescope Observing Program.



PLAYMATES Interacting pair NGC 4676, commonly known as "The Mice" because of the long tidal tails streaming from the galaxies, lies 300 million light-years away. This image, taken with the Hubble Space Telescope's Advanced Camera for Surveys, shows the charming yet catastrophic merger.

HERON GALAXY: DOUG MATTHEWS & E. J. JONES / ADAM BLOCK / NOAO / AURA / NSF; SKETCH: DALE HOLT; MICE GALAXY: NASA / HOLLAND FORD (JHU) / ACS SCIENCE TEAM / ESA



MID-WAY POINT The curled body of the Grasshopper, UGC 4881, shows two spiral galaxies halfway through a merger. The Orthoptera's body and head are particularly luminous, lit up by the process of star formation triggered by the galactic collision.

NGC 4676B is just 40″ southeast, connected to its neighbor by a diffuse tidal bridge. The luminous 30″ × 20″ halo, with a visual magnitude of 13.6, tilts northeast to southwest and increases to a vivid nucleus, but we saw only the initial extension of the southern tail.

The Grasshopper

Arp 55 (VV 155, UGC 4881) is a 14th-magnitude mid-stage merger of two gas-rich spirals, unusually luminous in the infrared (500 billion times the Sun's luminosity) because of intensive star formation in the nuclei and the tail. This tangled pair dwells 540 million light-years away in Lynx.

Vorontsov-Velyaminov assigned the nickname "Grasshopper" to this interacting pair because of the suggestive shape on the Palomar Observatory Sky Survey plates. In the eyepiece, the "head" and "body" appeared as a mottled, 30″ × 15″ glow. A nearly stellar knot on the southwest end marked the nucleus of the smaller galaxy, while the fused northeast member has a larger core but no discernible nucleus. The Grasshopper's "leg" is partially formed by a straight tail that extends 20″ due south from the northeast end. It brightened slightly at a star-forming site at the south tip.

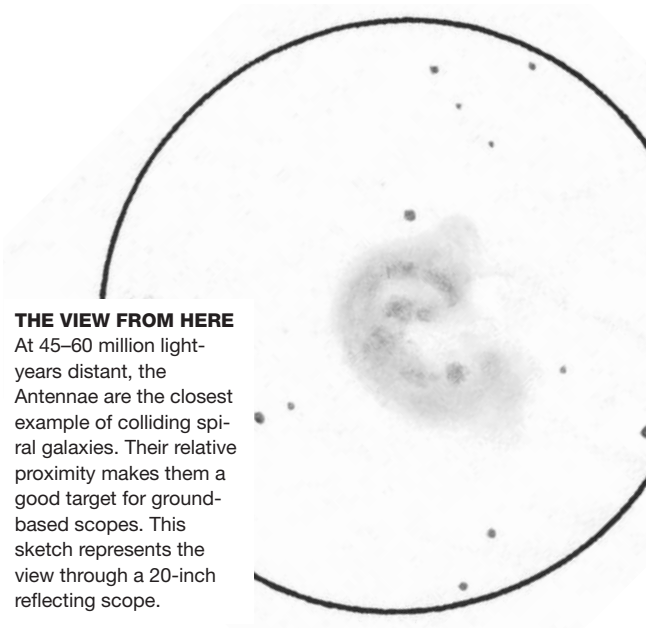
Three faint galaxies within 2.3′ of Arp 55 share the same redshift and form a small group. The closest is PGC 2242096. With a visual magnitude of 17.1, it appears as an amorphous 15″ patch just under 1′ northeast of the Grasshopper. Arp conjectured this object was another ejected "filament"; in a controversial 1967 paper he argued for a physical association between nearby disturbed galaxies and high redshift radio sources.

The Antennae

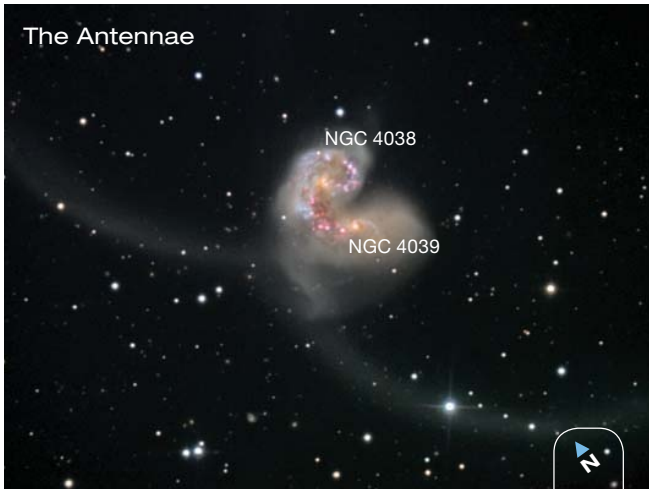
Arp 244 (VV 245) is made up of NGC 4038 and NGC 4039, galaxies found 3½° southwest of Gamma (γ) Corvi, the 2.6-magnitude star marking the northwest corner of Corvus's distinctive sail asterism. It's the closest example of an extreme mashup of two gas-rich spirals and really looks like colliding galaxies through an 8-inch telescope.

The Antennae galaxies provide an ideal astrophysical laboratory to investigate an ongoing interaction across all available wavelengths. At the galaxies' closest approach some 250 million years ago, gravitational tides pulled out two graceful narrow tails of gas, stars, and dust. Giant molecular clouds were compressed during the collision, triggering the formation of numerous super star clusters.

A few years ago I had an astonishing view of the Antennae using a 24-inch f/3.7 scope with the pair high overhead



THE VIEW FROM HERE At 45–60 million light-years distant, the Antennae are the closest example of colliding spiral galaxies. Their relative proximity makes them a good target for ground-based scopes. This sketch represents the view through a 20-inch reflecting scope.



EXTREME MASHUP The Antennae highlight the effects of galactic mergers. The collision triggered bursts of star production and the formation of numerous super star clusters. Graceful tidal tails, composed of stars, dust, and gas, arc away from the aftermath.

GRASSHOPPER GALAXY: NASA / ESA / HUBBLE HERITAGE TEAM (STSCI / AURA) / A. EVANS (UNIV. OF VIRGINIA / NRAO / STONY BROOK UNIV.); ANTENNAE GALAXY: NOAO / AURA / NSF / B. TWARDY / ADAM BLOCK; SKETCH: DALE HOLT

from Australia. NGC 4038 has a bizarre annular shape due to a distorted spiral arm that forms a bright circular rim. The off-center nucleus seems misplaced along the east portion of the rim, and sparkling clumps are embedded on the west and north border.

A shrimp-like tail emanates from the eastern end of NGC 4038 as a large curdled patch. The appendage hooks counterclockwise around the south side to NGC 4039's core, a prominent irregular nodule. A large, fainter envelope spreads further southwest and widens to a bulbous end.

I shared a memorable view through Lowrey's 48-inch last May, with *S&T* Contributing Editor Howard Banich. The two "antennae" were subtle and only the initial portions of the tails were easy, but the northern streamer included a mildly enhanced section. Tip-to-tip, the two ethereal wings spanned 10′. The "Wow!" moment came when Jimi added a narrow-band filter and numerous H II knots lit up along the rim. Howard compared the view to "looking at city lights from above on a foggy night."

Edward's Galaxy

NGC 6621 and NGC 6622 form **Arp 81** (VV 247), a unique interacting system with a single tidal tail that wraps along the east side of both galaxies. The merged duo is located in Draco, just 2.2° northeast of NGC 6543, the Cat's Eye planetary nebula.

So, who's Edward? He's the son of prolific comet hunter Lewis Swift. At the age of 14, Edward discovered NGC 6621 while assisting his father with observations. He's credited with the discovery of 48 NGC/IC objects between 1883 and 1891, but NGC 6621 is our favorite from that list.

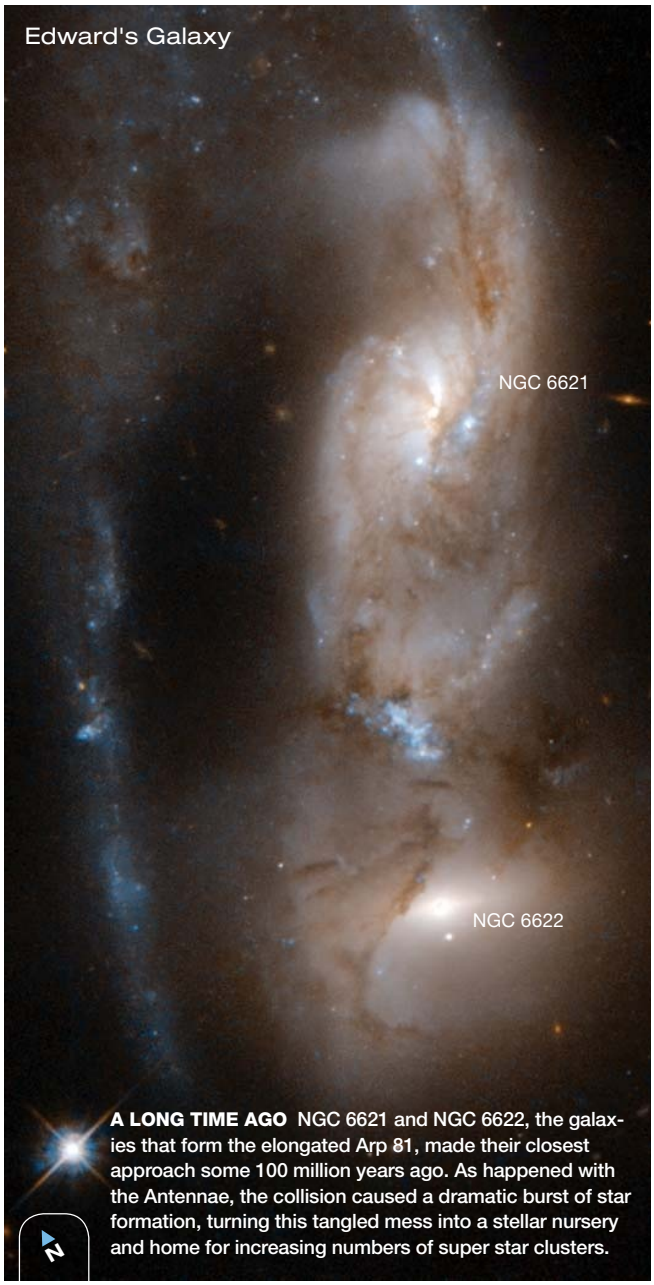
In 2003 astronomers William Keel (University of Alabama) and Kirk Bork (Institute for Science and Technology, Raytheon/NASA-Goddard) used observations from the HST Wide-Field Planetary Camera 2 and a computer *N*-body simulation to model Arp 81's one-sided tidal tail, the star-forming complexes bridging the region between the galaxy nuclei, and a twisting system of dust lanes. They found a near-grazing encounter of two equal-mass galaxies about 100 million years ago matched the current morphology.

With excellent seeing and transparency (Sky Quality Meter reading of 21.8), NGC 6621 appeared sharply concentrated at 697× with a brilliant 15″ core intensifying to the center. The main halo extended 1′ long and ⅓ as wide with slightly brighter "handles" marking the ends of the main spiral arms. The southeast end appeared fused with the halo of NGC 6622, creating a contiguous glow with the two cores separated by 40″. We glimpsed a brighter 10″ patch at the midpoint, which corresponds with a blue splash of starburst activity in the overlap region on the HST image.

The most startling feature we saw in the eyepiece, however, was the broad tidal arm that emerged to the north of NGC 6621 and folded back on itself nearly 180° to the south-southeast. The tail gradually faded while hugging the entire length of NGC 6621 and terminated in a small knot. A 15th-magnitude star sits 0.6′ east of NGC 6622, just beyond the end of the tail.

This view of Edward's Galaxy was a perfect ending to a thrilling night. Interacting galaxies offer an intriguing target, no matter the catalog, no matter the instrument. But in good seeing with a master-class scope, they become something truly spectacular to behold.

■ A dedicated observer, Contributing Editor **STEVE GOTTLIEB** usually keeps an eye on the sky for us from California. He welcomes your questions and comments at **steve_gottlieb@comcast.net**.



A LONG TIME AGO NGC 6621 and NGC 6622, the galaxies that form the elongated Arp 81, made their closest approach some 100 million years ago. As happened with the Antennae, the collision caused a dramatic burst of star formation, turning this tangled mess into a stellar nursery and home for increasing numbers of super star clusters.

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